Electrolyte Wetting Project: Engineering Blueprint

Oct 22, 2026 · @Matthew Sheets

1. Executive recommendation

Keep electrolyte wetting, but reframe the engineering question. A single pore through an electrode's thickness fills in milliseconds by Washburn's law, yet real cells take tens of minutes to wet. The project should explain that gap with fluid mechanics: wetting distance in a large cell is set by its in-plane size (the liquid enters from the edges), filling time grows with distance squared, and trapped gas resists filling unless the cell is evacuated first.

Engineering question: *How long does electrolyte take to wet an electrode from its edges, and how much do vacuum, pore size, contact angle and viscosity change that time and the fraction of pore space that ends up filled?*

This framing is supported by published neutron imaging of a prismatic cell, where electrolyte spread from all four edges inward, wetting finished in roughly 50 minutes under vacuum versus about 100 minutes at normal pressure, and about 10% less electrolyte was absorbed at normal pressure because of trapped gas (TUM/FRM II; TUM press release).

Baseline model: one idealized cylindrical capillary, solved in two clearly separate modes: (A) open capillary with capillary plus applied pressure, which has an exact analytical solution; (B) dead-end capillary with isothermal trapped gas, an ODE solved numerically and checked against its own exact equilibrium. Pore-network, Darcy/Kozeny–Carman and electrochemistry stay out of version 1.

Main revisions from your original idea: (1) the question shifts from "how fast does one pore fill" to "why does a cell take so long, and what does vacuum buy you"; (2) Darcy and Kozeny–Carman move from the core model to an optional extension, because they add parameters you cannot source consistently; (3) the experimental validation claim is downgraded to "order-of-magnitude plausibility", since no published dataset matches the model's geometry.

Status: go, conditional on four items in section 13. This blueprint can serve as your Model A plan for the rubric's two-model requirement; record that it came from Claude Opus 5.5 on Oct 22, 2026.

2. Topic comparison and final choice

Electrolyte wetting has the top total and wins on fluid-mechanics fit and educational value; its weakness is validation data. Scores are 1–5 and are my engineering judgment, not measurements.

Topic	Fluid-mech depth	Math feasibility	Originality	Validation data	Educational value	Implementation ease	Total /30
Electrolyte wetting (reframed)	5	5	4	3	5	5	**27**
Gas vessel blowdown (earlier top pick)	4	4	4	5	4	4	**25**
Compressible cake filtration	4	3	3	4	4	3	**21**
Paper wicking (same physics, no battery)	4	5	2	4	4	5	**24**
Marangoni flow (tears of wine)	4	2	4	3	4	2	**19**
Viscous fingering in porous media	4	2	4	2	4	2	**18**

Why wetting wins:

- **Fluid-mech depth:** surface tension, contact angle, viscous flow and a pressure balance are all core course topics, which your professor specifically asked you to include.
- **Feasibility:** the baseline has an exact solution, so verification is clean.
- **Educational value:** the "milliseconds vs. an hour" paradox and the vacuum effect are intuitive hooks.
- **Industrial relevance:** wetting is described as a time-consuming bottleneck in cell production, where manufacturers wait hours to ensure full absorption (TUM).

Honest trade-offs:

- **Cake filtration** has better published data (constant-pressure filtration tests) but needs a compressibility constitutive law that is empirical and harder to teach.
- **Blowdown** has stronger validation data but needs compressible flow, which your course may not reach.
- **Paper wicking** is the same physics with easier validation, but it is the least original and loses the battery story.

Final choice: electrolyte wetting, reframed. Paper wicking can appear as a "same physics elsewhere" example, not as the topic.

3. Governing equations and derivation roadmap

Five equations are kept; Darcy and Kozeny–Carman become optional. Every kept equation feeds the solver or a verification check.

Equation audit

Equation	Verdict	Role
Young's equation	Keep (concept only)	Defines contact angle θ; not computed
Young–Laplace capillary pressure	Keep	Driving pressure p_c
Hagen–Poiseuille	Keep	Viscous resistance behind the front
Washburn (with applied Δp)	Keep	Exact solution, Mode A; benchmark for Mode B
Isothermal ideal gas (Boyle)	Keep	Trapped-gas back-pressure, Mode B
Tortuosity scaling	Keep (labeled estimate)	Maps one pore to an electrode's straight-line distance
Darcy's law	Optional	Equivalent to Washburn for a capillary bundle; adds no new physics in v1

Equation	Verdict	Role
Kozeny–Carman	Optional	Needs particle size and an empirical constant; inconsistent with an effective pore radius

Conventions

θ is measured through the liquid. Spontaneous wetting requires θ < 90°. At θ = 90° the capillary drive is zero; above it, p_c is negative and the pore resists filling unless an applied pressure exceeds |p_c|.

Step-by-step derivation (baseline)

Step 1. Young's equation (balance of interfacial tensions at the contact line; σ is liquid–gas surface tension, σ_sg and σ_sl the solid–gas and solid–liquid interfacial tensions, all N/m):

```
\sigma_{sg} = \sigma_{sl} + \sigma\cos\theta
```

Step 2. Young–Laplace for a hemispherical-cap meniscus in a cylinder of radius r (geometry factor 2 is for a circular tube; a slit of half-width h gives σcosθ/h instead):

```
p_c = \frac{2\sigma\cos\theta}{r} \quad [\text{Pa}]
```

Step 3. Hagen–Poiseuille behind the front. Liquid occupies length L(t). Mean velocity for pressure drop Δp over length L, viscosity μ (Pa·s):

```
\frac{dL}{dt} = \frac{r^2\,\Delta p}{8\mu L}
```

Assumptions: fully developed laminar flow, no slip, Newtonian liquid, quasi-steady (inertia neglected), gravity neglected, meniscus volume and entrance losses neglected, static contact angle.

Step 4. Mode A, open capillary. Total drive Δp = p_c + Δp_a, where Δp_a is the applied pressure difference (liquid side minus gas side). Since Δp is constant, multiply by L and integrate with L(0) = 0:

```
L^2 = \frac{r^2\,(p_c + \Delta p_a)}{4\mu}\,t \qquad \xrightarrow{\Delta p_a
= 0} \qquad L^2 = \frac{r\,\sigma\cos\theta}{2\mu}\,t
```

The right-hand form is the Lucas–Washburn equation, the classical viscous-regime result (Fries & Dreyer, ZARM). Time to reach distance L:

$$t = \frac{4\mu L^2}{r^2 (p_c + \Delta p_a)}$$

Step 5. Mode B, dead-end capillary with trapped gas. A pore of length L_p is closed at its far end. Gas inside starts at absolute pressure p_g0 (atmospheric, or the vacuum level before filling) and is compressed isothermally as liquid enters. Liquid reservoir is at absolute pressure p_L.

$$p_g(L) = p_{g0}\,\frac{L_p}{L_p - L} \qquad \frac{dL}{dt} = \frac{r^2\,[\,p_L + p_c - p_g(L)\,]}{8\mu L}$$

Filling stops where the drive vanishes, giving an exact equilibrium you can check by hand:

$$\frac{L_{eq}}{L_p} = 1 - \frac{p_{g0}}{p_L + p_c}$$

This is a worst case: real pores are interconnected, so gas can escape through unfilled paths. Closed-capillary dynamics with gas compression have been studied in detail (Ramakrishnan, Wu, Zhang & Wasan, "Dynamics in closed and open capillaries", J. Fluid Mech., 2019; access it through UC's library).

Step 6. Tortuosity scaling (labeled estimate). If the liquid path is τ times longer than the straight-line distance X, then L = τX and:

$$X^2 = \frac{r\,\sigma\cos\theta}{2\mu\,\tau^2}\,t$$

This is the same result Darcy's law gives for a bundle of tortuous capillaries, which is why Darcy adds no new physics to v1.

Issues checked

- **Singularity:** dL/dt is infinite at L = 0. Solve for s = L² instead: ds/dt = r²Δp/(4μ), which is finite.
- **Inertia:** the viscous time scale ρr²/μ is about 7 × 10⁻⁸ s for r = 0.5 μm (ρ ≈ 1,300 kg/m³), so neglecting inertia is safe at pore scale.
- **Gravity:** Bond number ρgr²/σ is about 10⁻⁷, so gravity is negligible.
- **Invalid inputs:** r ⩽ 0, μ ⩽ 0, and L ⩾ L_p in Mode B are rejected. θ ⩾ 90° with Δp_a ⩽ |p_c| yields "no infiltration", not a negative time.

- **What the time means:** an idealized infiltration time for one representative pore, not a guaranteed cell filling time.

4. Baseline model specification

One idealized cylindrical capillary, two separate modes, one solver. Mode A is the analytical anchor; Mode B is the numerical contribution.

Why this over the alternatives: a Darcy porous-medium model would need permeability, porosity and an effective pore radius that you cannot source as one consistent set, and its answer reduces to Washburn anyway. A capillary-only model without trapped gas misses the vacuum effect, which is the most interesting engineering lever.

Item	Mode A: open capillary	Mode B: dead-end capillary with trapped gas
Inputs	r, σ, θ, μ, Δp_a, target distance X, tortuosity τ	r, σ, θ, μ, p_L, p_g0 (vacuum level), pore length L_p
Equations	Steps 2–4, 6	Steps 2, 3, 5
Method	Closed form, plus the same ODE solver for verification	ODE in $s = L^2$, adaptive RK45 (scipy solve_ivp) or fixed-step RK4
Outputs	L(t), time to X, p_c, Washburn coefficient	L(t), L_eq/L_p (fill fraction), t_{95} (time to 95% of L_eq), p_g(t)
Stop rule	L reaches X	L within 0.1% of L_eq, or a time cap

Fixed parameters: liquid density (only for the inertia/gravity validity checks), isothermal gas, T = 25 °C.

Plots: front position vs. time (with $\sqrt{t}$ reference line in Mode A); time to target vs. one swept input on log–log axes so the slope shows the exponent; fill fraction vs. vacuum level in Mode B.

Validity rules shown live: inertia time $\rho r^2/\mu \ll$ simulated time; Bond number $\ll 1$; Reynolds number of the front $\ll 1$; warning when $\theta > 80°$ (dynamic contact angle matters most there); warning when r falls below about 10 nm (continuum and slip assumptions weaken).

Limiting cases: $\theta \to 90°$ with $\Delta p_a = 0$ gives $t \to \infty$; $p_{g0} \to 0$ makes Mode B approach Mode A; $L_p \to \infty$ also recovers Mode A; doubling μ doubles time; doubling r halves time in pure capillary filling.

Optional extensions (after v1 works): dynamic contact angle; Kozeny–Carman estimate of r from particle size and porosity; gas dissolution; filling from both ends.

5. Verification and validation plan

Verification asks whether the code solves the equations correctly; you can fully achieve it. **Validation** asks whether the equations describe real electrodes; you can only partly achieve it, and the site must say so.

A. Analytical verification

Run the Mode B ODE solver with p_g0 = 0 (or L_p very large). It must reproduce the Mode A closed form $L^2 = r^2(p_c + \Delta p_a)t/(4\mu)$. Pass: relative error below 10^{-6} at all output times. Second check: Mode B's long-time L must match the exact L_eq to within 0.1%.

B. Independent hand calculation

Property values: an EC/DMC 50:50 electrolyte with $LiPF_6$ (0.787 mol/kg) measured σ = 29.4 mN/m, μ = 4.4 mPa·s at 20 °C, density 1.303 g/cm³ (KIT, Int. J. Mol. Sci. 2015). Contact angle 30° and pore radius 0.5 μm are **assumptions** to be sourced (section 12).

1. $p_c = 2(0.0294)(\cos 30°)/(0.5 \times 10^{-6}) =$ **101.8 kPa**, about one atmosphere.
2. Washburn coefficient $r\sigma\cos\theta/(2\mu) =$ **1.447×10^{-6} m²/s**.
3. Through a 70 μm electrode thickness: $t = (70 \times 10^{-6})^2/1.447 \times 10^{-6} =$ **3.4 ms**.
4. In-plane over 1 cm: **69 s**. Over 5 cm: **1,728 s ≈ 29 min**. With tortuosity τ = 2 (time × 4): **≈ 115 min**.
5. Mode B, liquid at 101.3 kPa: ambient start (p_g0 = 101.3 kPa) gives fill fraction **0.50**; 10 kPa vacuum gives **0.95**; 1 kPa gives **0.995**.

Teaching point: the 5 cm estimate is the same order of magnitude as the ~50–100 minute wetting times reported for a prismatic cell (TUM). That is a plausibility check, not validation: the cell's geometry, pore size and contact angle are unknown. The dead-end model's 50% trapping at ambient pressure is far worse than the ~10% lower uptake observed, which shows gas escapes through connected pores. Present that gap honestly; it is a good discussion point.

C. Dimensional analysis

- p_c: $(N/m)/m = N/m^2 = Pa$ ✓

- $r^2\Delta p/(\mu L)$: $m^2 \cdot Pa/(Pa \cdot s \cdot m) = m/s$ ✓
- Washburn coefficient: $m \cdot (N/m)/(Pa \cdot s) = m^2/s$ ✓
- p_g: Pa × (m/m) = Pa ✓

D. Limiting-case tests

Change	Expected (from equations)	Where model fails
μ × 2	time × 2	Non-Newtonian or temperature-dependent liquids
r × 2, $\Delta p_a = 0$	time ÷ 2	Very small r: slip, surface roughness; very large r: gravity, inertia
θ → 90°	time → ∞	Dynamic contact angle differs from static
θ > 90°, $\Delta p_a < \lvert p_c \rvert$	no filling	Real surfaces show hysteresis
$p_{g0} \to 0$	Mode B → Mode A	Gas dissolution, vapor pressure of solvent
Distance × 2	time × 4	Non-uniform pores, edge effects

E. Numerical accuracy

Compare solutions at tolerance 10^{-6}, 10^{-8}, 10^{-10} (or step sizes h, h/2, h/4 for RK4). RK4 error should drop about 16× per halving. Report t_{95} change between the two tightest runs. Check that the solver handles the near-equilibrium slowdown without stalling.

F. Physical validation (what exists, and its limits)

Source	What was measured	Usable for	Mismatch
Davoodabadi et al. 2019, J. Power Sources	Electrolyte imbibition rates into electrodes; higher salt concentration slowed wetting; graphite anode wetted faster than NMC532 cathode	Trend check: $\sqrt{t}$ behavior, viscosity effect	Real electrodes, not single pores; paper may be paywalled
Weydanz et al. 2018, via TUM	Neutron images of a prismatic cell: edge-inward wetting, ~50 min vacuum vs ~100 min ambient	Qualitative vacuum effect and edge-wetting framing	No pore-scale parameters
Fries & Dreyer capillary-rise work	Capillary rise in tubes compared with Lucas–Washburn	Validates the Washburn physics itself	Glass tubes, not electrodes

Claim you can make: the code is verified; the physics is consistent with published capillary experiments and with observed trends in electrodes. **Claim you cannot make:** that the model predicts a specific cell's filling time.

6. Final teaching outline

Nine essential sections, each feeding the next. Every equation appears where it is first needed by the model.

1. **The paradox** (hook)
   - Objective: see why wetting is an engineering problem.
   - Concept: one pore fills in milliseconds; a cell takes about an hour.
   - Calculation: the 3.4 ms vs. ~29 min numbers from section 5.
   - Visual: cell cross-section with liquid entering from four edges.
   - Evidence: TUM/Bosch neutron imaging.
   - Next: what pulls liquid into a pore at all?

2. **Surface tension and interfacial tension**
    - Objective: explain where the pulling force comes from.
    - Equation: Young's equation (concept, not computed).
    - Visual: droplet with three tension arrows; slider for θ that redraws the drop.
    - Next: turning that pull into a pressure.
3. **Contact angle and wetting**
    - Objective: predict wet vs. non-wet from θ.
    - Concept: θ < 90° wets, θ > 90° resists; separator angles in the literature span roughly 0–65°.
    - Visual: cosθ curve crossing zero at 90°.
    - Next: geometry turns surface tension into pressure.
4. **Capillary pressure**
    - Objective: compute p_c for a pore.
    - Equation: Young–Laplace, $p_c = 2\sigma\cos\theta/r$.
    - Relevance: the driving term in every model mode.
    - Evidence: hand calculation, 101.8 kPa.
    - Next: what resists the flow?
5. **Viscous resistance in a pore**
    - Objective: connect to Hagen–Poiseuille from class.
    - Equation: $dL/dt = r^2\Delta p/(8\mu L)$.
    - Visual: parabolic velocity profile behind a moving meniscus.
    - Next: balance the two.
6. **The Washburn equation**
    - Objective: derive $L \propto \sqrt{t}$ and the time-to-distance formula.
    - Visual: front position vs. t and vs. $\sqrt{t}$ (straight line).
    - Evidence: analytical benchmark; classical capillary-rise experiments.
    - Next: why does distance matter so much?
7. **From one pore to a cell**
    - Objective: explain the paradox with L^2 scaling and tortuosity.
    - Equation: $X^2 = r\sigma\cos\theta\, t/(2\mu\tau^2)$.
    - Visual: interactive slider for distance from 70 µm to 10 cm on a log axis.
    - Next: what slows real filling further?

8. **Trapped gas and vacuum filling**
    - Objective: explain why manufacturers evacuate cells.
    - Equations: Boyle's law back-pressure; exact L_eq.
    - Visual: animated pore with gas bubble compressing; fill fraction vs. vacuum level.
    - Evidence: ~50 vs. ~100 min observation; honest gap discussion (50% predicted trapping vs. ~10% observed).
    - Next: how far can we trust this?
9. **Limits and the same physics elsewhere**
    - Objective: state what the model can and cannot predict.
    - Content: validity checks, dynamic contact angle, connected pores, paper and coatings examples.

Optional advanced extensions: dynamic contact angle; Kozeny–Carman pore-size estimate; Darcy formulation; inertial (Bosanquet) start-up regime.

Excluded from v1: pore-network or CFD simulation, electrochemistry and SEI formation, gas dissolution kinetics, non-Newtonian electrolytes.

7. Final website architecture

Seven tabs on one static site hosted on GitHub Pages, with the solver in Python running in the browser through Pyodide (Python compiled to WebAssembly). That meets the recommended WebAssembly approach while keeping all code in a language you can read and defend.

Lessons from the reference site

Your professor's packed-bed runaway site pairs a sidebar of inputs with an app tab, a numbered theory section, a step-by-step Python walkthrough that re-runs live in the browser, and a fast Rust/WebAssembly solver cross-checked against Python and SciPy, with tests against an exact Bessel-series solution.

- **Adopt:** sidebar inputs with presets; theory in numbered sections; step-by-step code cells with printed outputs; an independent solver cross-check; explicit tests against an exact solution.
- **Improve for intro students:** start from a physical hook instead of the full model; define every symbol on first use; add diagrams and a guided challenge; keep code walkthrough shorter.

- **Skip:** the Rust port and performance race. Your model solves in milliseconds, so speed is not the story; correctness and teaching are.
- **Add (not on the reference site):** the rubric items it does not need to show, namely video, transcript, AI process, model comparison and contributions.

Tabs

Tab	Purpose	Content	Rubric items
1. Overview	Hook and framing	Paradox, engineering question, what the tool predicts, AI-project statement, embedded video	1, 2, 4
2. Lab	Interactive model	Sidebar inputs, Mode A/B toggle, plots, validity warnings, guided challenge	1, 4
3. Theory	Teach the physics	Sections 1–9 of the teaching outline, diagrams, derivations	4
4. Code & Verification	Show the computation is right	Step-by-step Python cells; analytical benchmark; hand calc; convergence; limiting cases; published-data comparison; limitations	4
5. AI Lab	Show the AI process	Model A plan, Model B critique, decision table, prompts log, two implementations compared on shared tests	4, 5
6. Presentation	Required media	YouTube link, verbatim auto-transcript, note pointing to practice transcript and AI critique in YouTube comments, what you learned, future questions	2, 3, 4
7. References & Credits	Attribution	Scientific sources, software, AI models and dates, professor's site as design inspiration, contributions	6

The quiz sits at the end of Theory (tab 3), with feedback explaining each wrong answer, rather than its own tab.

8. Interactive laboratory specification

Eight inputs, three plots, one guided challenge. Ranges marked "assumption" must be sourced or labeled as illustrative on the site.

Input	Units	Default	Range	Basis	Effect	Invalid / warning
Pore radius r	µm	0.5	0.05–5	Assumption (to source)	$p_c \propto 1/r$; resistance $\propto 1/r^2$; net time $\propto 1/r$	⩽ 0 rejected; < 0.01 warns
Surface tension σ	mN/m	29.4	20–40	KIT measured 23.3–38.1	$p_c \propto \sigma$	⩽ 0 rejected
Contact angle θ	°	30	0–120	Separator values ~0–65° reported	$p_c \propto \cos\theta$	> 80° warns; ⩾ 90° may give no filling
Viscosity µ	mPa·s	4.4	1–10	KIT; patents cite ⩽ 5 mPa·s carbonate electrolytes	time $\propto \mu$	⩽ 0 rejected
Applied Δp (Mode A)	kPa	0	0–200	Assumption	adds to p_c	negative allowed only if total drive > 0
Distance X	µm to cm (log slider)	1 cm	10 µm–10 cm	Geometry	time $\propto X^2$	⩽ 0 rejected
Tortuosity τ	–	1	1–5	Assumption; labeled estimate	time $\propto \tau^2$	< 1 rejected
Initial gas pressure p_{g0} (Mode B)	kPa abs	101.3	0.1–101.3	Vacuum level	sets L_{eq}	> $p_L + p_c$ means no filling

Liquid-side pressure p_L is fixed at 101.3 kPa in Mode B, with a note explaining the choice.

Outputs (with units): p_c (kPa); time to target (s, min or h, auto-scaled); Washburn coefficient (m^2/s); fill fraction L_{eq}/L_p; t_{95}; validity-check badges (inertia, Bond number, θ warning).

Plots that earn their place:

1. Front position vs. time, numerical points over the analytical curve (shows verification live).
2. Time to target vs. a chosen input on log–log axes (the slope reveals the scaling exponent: 2 for distance, 1 for viscosity, −1 for radius).
3. Mode B: fill fraction vs. vacuum level, with the exact L_eq curve and ODE end points.

Guided challenge: "A cell maker wants electrolyte to travel 4 cm in under 30 minutes. With σ, μ and θ fixed at the defaults, what is the smallest pore radius that works? If they switch to a more viscous electrolyte (μ = 8 mPa·s), what must change? Finally, what vacuum level keeps trapped gas below 5% in a dead-end pore, and why might real cells do better than this model?" Students answer in a text box; the site reveals the worked solution and the reasoning.

9. Two-model AI workflow and evaluation framework

Treat the AI comparison like a lab experiment: same specification, same test cases, same scoring, evidence saved at every step.

Workflow and evidence to keep

1. **Model A plan:** this blueprint (Claude Opus 5.5, Oct 22, 2026). Save it as a PDF and save the exact prompt you used.
2. **Model B critique:** give a different model (from another company, so its critique is independent) this blueprint plus the assignment requirements. Ask it to find errors, missing items and over-complications. Save the full exchange.
3. **Your decisions:** fill the comparison table below for every suggestion. Check disputed equations by hand yourself.
4. **Frozen spec:** the approved version of sections 3, 4, 5 and 8. Both implementations must build to this, unchanged.
5. **Two implementations:** prompt each model separately with the frozen spec to write the solver and one page of the lab. Use identical prompts.
6. **Shared test suite:** written by you (with help) before either implementation exists: the analytical benchmark, the hand-calculation values, limiting cases, invalid inputs, convergence.
7. **Results:** run both implementations against the suite; record pass/fail and numbers.
8. **Your corrections:** log every bug you fixed by hand and what you learned from it.

Fairness rules

- Same frozen spec, same prompt wording, same test inputs.
- Count the number of follow-up prompts each model needed; don't give one model extra hints.
- Record model names, versions and dates.
- Report failures as findings, not embarrassments.

Scoring criteria (score each 0–2 per model)

Equation and unit correctness · analytical benchmark agreement · numerical robustness (near equilibrium, extreme inputs) · input validation · explanation quality · educational usability · code clarity · compliance with the rubric · follow-up prompts needed.

Comparison table template

Issue or feature	Model A proposal	Model B critique or implementation	Independent evidence	My decision	Reason	Test outcome
Example: drop Kozeny–Carman from v1	Optional only	(fill in)	Parameter-consistency argument, section 4	(accept / reject / modify)	(your reasoning)	(n/a or test result)

10. Rubric compliance matrix

Every item is **planned, not complete**. Each is done only when its acceptance test passes.

Requirement	Location	Evidence needed	Your personal work	Depends on	Common failure	Acceptance test
1. Public website with demo	GitHub Pages, tabs 1–2	Live URL; working lab	Repo setup, deploy, final review	Frozen spec, solver	Repo private; Pyodide fails to load on some browsers	Opens in a logged-out private window on phone and laptop
2. YouTube video linked	Tabs 1 and 6	Public or unlisted video link	Record presentation	Slides, finished site	Video set to private	Link plays when logged out

Requirement	Location	Evidence needed	Your personal work	Depends on	Common failure	Acceptance test
2. Practice transcript + AI critique in YouTube comments	YouTube comments	Pasted comment(s); critique prompt included project link and requirements	Record practice run, request critique, paste	Practice recording	Prompt omits project link or rubric; comment too long (split it)	Comment visible when logged out
3. Verbatim auto-transcript	Tab 6	Unedited YouTube auto-transcript text	Copy after upload	Final video processed	Cleaned-up or paraphrased transcript	Text matches YouTube's transcript panel exactly
4. Documentation: AI, process, reasoning, physics, learning	Tabs 3–6	Model list, prompts, iteration log, decisions, theory, reflection	Write reasoning and reflection yourself	All prior work	Generic AI-written reflection	Each decision has a stated reason; reflection names specific things learned
5. Model A plan, Model B critique, accept/reject reasoning	Tab 5	Saved plan, critique, filled decision table	Every accept/reject decision	This blueprint	Critique pasted without decisions	Table has a decision and reason per suggestion
5. Two-model implementation compared	Tab 5	Both codebases, shared test results	Write test suite; run tests	Frozen spec	Different specs or prompts for each model	Same suite run on both; results table published
6. References and attribution	Tab 7	Scientific, software, AI and design-inspiration citations	Verify every link	All content	Broken or invented citations	Every link opens; every number on site traces to a source or is labeled an assumption
6. Contributions	Tab 7	Statement of who did what (individual: you did all)	Write it	–	Omitted for individual work	Present and specific

Requirement	Location	Evidence needed	Your personal work	Depends on	Common failure	Acceptance test
Learning reflection and future work	Tab 6	What you learned, what you'd study next	Write it	Finished project	Vague statements	At least three concrete technical items

11. Prioritized implementation roadmap

Physics and tests come before any website code; the AI comparison is built into the order, not added at the end.

1. **Save evidence now:** this blueprint, the prompt that produced it, and your earlier topic-selection chats.
2. **Get Model B's critique** of this blueprint; fill the decision table.
3. **Resolve the open items** in section 12 (pore radius source, contact angle source, Davoodabadi access).
4. **Re-derive sections 3 and 5 by hand** on paper. Scan or photograph them for the site.
5. **Freeze the spec** (sections 3, 4, 5, 8 after revisions).
6. **Write the shared test suite** in Python (pytest) from section 5.
7. **Implementation 1** with Model A from the frozen spec; run tests; log fixes.
8. **Implementation 2** with Model B, same prompt; run tests; log fixes.
9. **Compare and choose** the solver for the final site (or merge the better parts); document why.
10. **Build the site** tabs in order: Lab → Theory → Code & Verification → AI Lab → Overview → References.
11. **Practice presentation**, get the AI critique (prompt includes site link and requirements), revise.
12. **Record the final video**, upload, wait for the auto-transcript, paste it verbatim into tab 6, post practice transcript and critique as comments.
13. **Run every acceptance test** in section 10 from a logged-out browser.

12. Stress test: outstanding risks and questions

Reviewed as a skeptical professor would. Items marked **open** need research, data or testing before they are resolved.

Five biggest technical risks

1. **Pore radius is unsourced (open).** Every time prediction scales with 1/r. Find a published pore-size distribution for a calendered graphite or NMC electrode (mercury porosimetry studies) and cite it, or present r only as a swept variable.
2. **"Effective" radius is not one number.** Real electrodes have a pore-size distribution; capillary pressure favors small pores, flow favors large ones. State that a single r is a modeling choice.
3. **Contact angle is uncertain (open).** Static angles on separators range widely, and advancing (dynamic) angles are higher than static. Cite the source values used and show sensitivity.
4. **Trapped-gas model overpredicts trapping.** Dead-end pores give 50% trapping at ambient pressure; observations suggest far less. Present Mode B as an upper bound and explain why.
5. **Validation is qualitative.** No dataset matches the model geometry. If Davoodabadi et al. is accessible through UC's library, digitize their imbibition curves and test the $\sqrt{t}$ trend and the salt-concentration (viscosity) effect.

Unsupported assumptions to label on the site

$r = 0.5\ \mu m$; $\theta = 30°$; $\tau = 1–5$; $p_L = 101.3\ kPa$; isothermal gas; the in-plane wetting distance of a real cell.

Equations that could be misused

- Kozeny–Carman with an inconsistent pore radius (kept out of v1 for this reason).
- Washburn applied to the whole cell without the τ and pore-distribution caveats.
- Young's equation presented as something the model solves (it only defines θ).

Complexity without educational value (cut)

A Rust/WebAssembly port and performance race; 2D CFD of the meniscus; a full pore network.

Easy-to-miss website requirements

Verbatim (unedited) transcript; critique prompt that includes the site link and requirements; comments visible publicly; model versions and dates; contributions statement even for individual work.

Why it could still score poorly

If the site reads as AI-generated without your reasoning; if numbers lack sources; if the AI comparison uses different specs per model; if you can't re-derive Washburn live during questions.

Changes that most improve quality

Sourcing r and θ; digitized published imbibition data; your hand derivations shown on the site; an honest "model vs. reality" panel.

13. Go/no-go assessment

Go, conditionally. The physics, model, tests and site structure are ready to freeze. Four items must be resolved before any implementation code is written:

- [ ] Model B critique received and every suggestion decided in the comparison table.
- [ ] A cited source (or explicit "illustrative" label) for pore radius and contact angle.
- [ ] You have re-derived the Washburn and trapped-gas equations by hand and reproduced the section 5 numbers.
- [ ] Shared test suite written and agreed before either model writes the solver.

Optional but valuable: library access to Davoodabadi et al. (2019) for quantitative trend validation.

Sources

Searched and read from search excerpts unless noted; open each before citing on your site.

- Filling lithium-ion cells faster — TUM / FRM II (Weydanz et al., J. Power Sources 380, 2018)
- Neutrons pave the way to accelerated production of lithium-ion cells — TUM press release

- In situ visualization of the electrolyte solvent filling process by neutron radiography — J. Power Sources 331 (2016)
- Davoodabadi et al., Analysis of electrolyte imbibition through lithium-ion battery electrodes — J. Power Sources 424 (2019), ORNL record
- Electrolyte properties incl. surface tension 29.4 mN/m and viscosity 4.4 mPa·s — Int. J. Mol. Sci. 16 (2015), KIT repository
- Separator contact angle 65.4° with 1 M $LiPF_6$ EC/DMC — Nano-Micro Letters (2021)
- Treated battery separator patent: contact angles 26.3° (untreated) and 8.3° (treated) — USPTO 9083034
- Analytic solutions of capillary rise — Fries & Dreyer, ZARM Bremen
- Fries & Dreyer, Dimensionless scaling methods for capillary rise — J. Colloid Interface Sci. 338 (2009), preprint
- Ramakrishnan, Wu, Zhang & Wasan, Dynamics in closed and open capillaries — J. Fluid Mech. (2019); access through UC library
- Wetting of battery electrodes (Washburn method, filling-time bottleneck) — KRÜSS
- Packed-Bed Reactor — Hot Spots & Runaway (professor's reference site), opened in full